These instructions worked on October 11th 2023

Load relevant modules:

You can choose whether to se GNU or Intel modules, here we use GNU.

You can load modules by hand every time, or setup a script with the following lines.

```
module load cray-fftw
```

Compile libconfig

Clone the github repository for libconfig: https://github.com/hyperrealm/libconfig, and type

```
mkdir $HOME/Install.cori
cd libconfig
autoreconf
./configure --prefix=$HOME/Install.perlmutter
make MAKEINFO=true
make install MAKEINFO=true
```

Compile libsharp

Clone the github repository for libsharp: https://github.com/Libsharp/libsharp, and type:

```
cd libsharp
autoreconf
./configure
make
```

Compile cfitsio

Download and unpack from http://heasarc.gsfc.nasa.gov/FTP/software/fitsio/c/cfitsio-4.2.0.tar.gz:

```
./configure
make
make install
```

Compile healpix

Download and unpack from https://healpix.sourceforge.io/documentation.php (``wget https://sourceforge.net/projects/healpix/files/latest/download -O healpix.zip``)

```
./configure #And follow instructions to install C package
make
```

Environmental variables

Make sure that the libraries used to compile `CoLoRe` are in `LD_LIBRARY_PATH` so they can be found at runtime:

```
export CFITSIO_DIR=/global/common/software/desi/users/cramirez/cfitsio-4.2.0
export HEALPIX_DIR=/global/common/software/desi/users/cramirez/Healpix_3.82
export LIBSHARP_DIR=/global/common/software/desi/users/cramirez/libsharp
export LIBCONFIG_DIR=/global/common/software/desi/users/cramirez/Install.perlmutter
export FFTW_DIR=/opt/cray/pe/fftw/3.3.10.3/x86_rome

export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/usr/lib64:${FFTW_DIR}/lib:${CFITSIO_DIR}/lib:${LIBSHARP_DIR}/auto/lib:${LIBCONFIG_DIR}/lib
```

Modify CoLoRe's makefile

```
COMP_SER = cc
COMP_MPI = cc
OPTIONS = -Wall -O3 -std=c99

DEFINEFLAGS += -D_LONGIDS
DEFINEFLAGS += -D_BIAS_MODEL_3 #
DEFINEFLAGS += -D_DEBUG

## Using single precision reduces the memory footprint
USE_SINGLE_PRECISION = yes

ADD_EXTRA_KAPPA = no
USE_HDF5 = no
USE_FITS = yes
USE_OMP = yes
USE_MPI = yes
```

```
GSL_INC = -I$/usr/include
GSL_LIB =  -L/usr/lib64 -lgsl -lgslcblas

FFTW_INC = -I${FFTW_DIR}/include
FFTW_LIB = -L${FFTW_DIR}/lib

FITS_INC = -I${CFITSIO_DIR}/include
FITS_LIB = -L${CFITSIO_DIR}/lib -lcfitsio

HDF5_INC =
HDF5_LIB =

CONF_INC = -I${LIBCONFIG_DIR}/include
CONF_LIB = -L${LIBCONFIG_DIR}/lib

HPIX_INC = -I${HEALPIX_DIR}/include
HPIX_LIB = -L${HEALPIX_DIR}/lib

SHT_INC = -I${LIBSHARP_DIR}/include
SHT_LIB = -L${LIBSHARP_DIR}/lib
```